Appendices:
Cheap Metrics First, Replanning Diagnostics Second

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Appendix A
Frozen Analysis Contract

The analysis unit is a task–dataset–policy–seed training run. Checkpoint correlations are computed within runs, averaged over runs within each task, then averaged with equal task weight. Task-clustered bootstrap intervals resample tasks and nested runs. Outcome-resolving runs require success IQR at least 0.10, at least five distinct success values, and a best–worst success gap at least 0.20.

Regret@q retains $\lceil qN \rceil$ checkpoints; Regret@1 retains exactly one. Lower-cost metrics sort first; boundary ties use checkpoint path in the frozen manifest, never rollout success. Each primary checkpoint has 50 simulator rollouts (0.02 success increments).

SIMPLER's MMRV averages each checkpoint's largest true-success margin to a checkpoint incorrectly ordered by the metric, estimating margins from rollout rates. Lower is better; large reversals count more than unresolved near-ties. Primary task-balanced values are 0.153 for Action L1, 0.160 for PLV-1 smoothness, 0.166 for open-loop smoothness, and 0.290 for objective loss.

Appendix B
Policy-Family Split and Design Resolution

With five task means, paired differences detectable at 80% power are: PLV-1 smoothness versus Action L1 (observed $\Delta\rho = 0.016$, detectable $\Delta\rho \approx 0.117$); PLV-1 smoothness versus open-loop smoothness (observed $\Delta\rho = 0.019$, detectable $\Delta\rho \approx 0.101$); open-loop smoothness versus Action L1 (observed $\Delta\rho = -0.003$, detectable $\Delta\rho \approx 0.073$). No observed gap reaches 0.19 of its detectable difference. Comparisons are unresolved, not proven equivalent; the inference unit is the task, not the checkpoint.

All metric–success relations show curvature: a quadratic term raises R^2 by 0.15–0.19 over a linear fit, with consistently negative quadratic coefficient; LOWESS averages 1.4–1.7 turning points per run. We therefore summarize by ranks; linear fits overstate late-training gains.

Action-space and PLV smoothness signals have mean within-run pairwise rank correlations 0.53–0.66; the first rank component explains 73.9% of variance. PLV-1's partial rank association after Action L1 and open-loop smoothness is 0.183: remaining association, not an established selection gain.

Appendix C
Score Normalization Sensitivity

Normalization is pool-dependent, not intrinsically calibrated. Task-balanced Spearman is 0.366 with target-pool median/IQR, 0.364 with a fixed seed reference, 0.321 with within-run empirical ranks, and 0.178 without normalization. Random half-pools retain ordering ($\rho = 0.961$) but shift scores by 0.186 on average. Score magnitudes are interpreted only within the declared pool.

Appendix D
Finite-Rollout Reliability and Baselines

A binomial bootstrap resamples outcomes at each checkpoint's observed rate, preserving the run/task hierarchy.

The primary archive stores success counts, not episode order. Conditional on each 50-episode count, we draw one 25-episode half from the exact hypergeometric distribution and assign the remaining successes to the other. Over 1,000 repetitions, task-balanced split-half Spearman is 0.500 with interval $[0.472, 0.529]$; Spearman–Brown reliability for 50 outcomes is 0.667. This diagnoses conditional reliability, not correlation with an unobserved true success probability.

Checkpoint i with k_i successes receives a Beta$(k_i + 1, 50 - k_i + 1)$ posterior. Pair ordering is scored only when $P(S_i > S_j) > 0.95$ or its reverse exceeds 0.95; adjacent observed rates such as 0.48 and 0.50 do not establish certain performance differences.

At 30%, random mean regret is 0.046 with randomization interval $[0.026, 0.064]$; selecting the last checkpoint gives 0.118. Table V reports other baselines. With only five nonzero task differences, the smallest attainable exact two-sided signed-rank p-value is 0.0625.

TABLE I

PRIMARY POLICY-FAMILY SPLIT: PLV-1 LEADS ON DIFFUSION POLICY, OPEN-LOOP SMOOTHNESS ON BC-RNN. OBJECTIVE LOSS IS ADVERSE ON DP BUT COMPETITIVE ON BC-RNN. DESCRIPTIVE FIVE-TASK SUBGROUP ESTIMATES; NO MULTIPLICITY CORRECTION. ARROWS GIVE PREFERRED DIRECTIONS.

Metric	$\rho \uparrow$	MMRV$\downarrow$	Regret@30$\downarrow$
BC-RNN (21 runs)			
Action L1	0.372	0.190	0.050
Open-loop smoothness	0.401	0.184	0.028
PLV-1 smoothness	0.303	0.231	0.084
PLV-1 uncertainty+smooth	0.344	0.224	0.076
Objective validation loss	0.360	0.261	0.034
Diffusion Policy (24 runs)			
Action L1	0.462	0.120	0.017
Open-loop smoothness	0.418	0.149	0.010
PLV-1 smoothness	0.556	0.093	0.006
PLV-1 uncertainty+smooth	0.502	0.102	0.008
Objective validation loss	-0.243	0.316	0.083

APPENDIX E
COHORTS AND EVIDENCE ROLES

a) External blinding deviation: Strict prospective external evaluations are preregistered, not fully analyst-blind. Methods, budgets, gates, confirmatory analysis, and reporting code were frozen before target outcomes. A filesystem inventory printed validation scores embedded in checkpoint filenames; a live integrity audit exposed normalized partial sparse rewards for one task before aggregate ranking analysis. No episode-level outcomes, registered confirmatory decisions, all-budget endpoint reports, or paired bootstrap results were exposed. The selector script and thresholds were already fixed; registered analysis remained unchanged. Filename scores and intermediate rewards were not fully embargoed.

APPENDIX F
DIFFUSION POLICY EXECUTION CONTRACT

The low-dimensional Diffusion Policy uses prediction horizon 16, observation horizon two, executable offset $s = 1$, stride $m = 8$, and absolute actions. The committed slice is $A_t[s : s + m]$. Action L1 compares the sampled executable slice with the aligned expert slice. The global metric uses one sampled evaluation; execution controls use four branch-matched diffusion seeds and take their median after averaging validation starts.

Teacher, model, and simulator branches share start, old plan, stride, and sampling seed. At the next replan, the teacher reads demonstration history; the model shifts history by one observation per predicted transition. The simulator restores the demonstration simulator-state snapshot, executes the same committed absolute-action slice converted to the environment convention, and appends resulting observations.

With four policy seeds and distance d between aligned plans, we separate diffusion stochasticity:

$$J_{\text{sample}} = d(\pi(x;\xi_1), \pi(x;\xi_2)), \tag{1}$$

$$J_{\text{state}} = d(\pi(x;\xi), \pi(x';\xi)), \tag{2}$$

$$J_{\text{total}} = d(\pi(x;\xi_1), \pi(x';\xi_2)). \tag{3}$$

These change sampling only, state only (common random numbers), or both (stochastic deployment), respectively. Common random numbers reuse the diffusion sampling seed. High sampling-only distance indicates policy stochasticity, not closed-loop state sensitivity.

The clean mechanism contrasts are

$$D_{M,E} = d(A^M, A^E), \qquad D_{S,E} = d(A^S, A^E), \tag{4}$$

$$D_{M,S} = d(A^M, A^S). \tag{5}$$

The completed 47-checkpoint archive stored only branch distances to the old-plan suffix, supporting the weaker excess proxy $\Delta J^M = J^M - J^E$, not reconstruction of direct contrasts. The evaluator now records direct

TABLE II

COHORT MAP. COUNTS ARE NON-ADDITIVE: THE 504, 130, AND 47 ROWS ARE NESTED PRIMARY-POOL SUBSETS. SELECTED GRIDS ARE NOT COMPLETE-GRID CONFIRMATIONS.

Cohort	Policy	Tasks/ runs	Ckpts	Rollouts/ ckpt	Grid	Evidence role
Primary retrospective	BC-RNN, DP	5/45	1,365	50	heterogeneous	main analysis
Execution archive	DP	5/24	504	50	archived	development
Four-seed controls	DP	5/24	130	50	retained	post-hoc control
Same-start branches	DP	4/9	47	50	retained	mechanism
Coffee/StackThree	BC-RNN	2/8	328	50	complete	frozen
Eight-task breadth	BC-RNN	8/32	1,312	50	complete	frozen
Action-chunk breadth	DP	8/8	168	50	complete	frozen
Unseen family	ACT	2/6	126	50	complete	transfer
Domain variants	ACT	2/6	126	50	complete	strict prospective
Unseen semantic family	ACT	2/6	126	50	complete	strict prospective
Strict sequence	ACT	17/51	561	50	complete	prospective
Candidate-only stress	ACT	3/9	40	50	selected	non-discriminative

TABLE III

EXTERNAL EVIDENCE. "CONFIRM." REQUIRES FROZEN, COMPLETE, OUTCOME-RESOLVING COVERAGE; "DESCRIPTIVE" MEANS SELECTED OR INCOMPLETE COVERAGE; "FAILED" MEANS AN UNCONFIRMED FROZEN PLV CLAIM; "N/D" MEANS NON-DISCRIMINATIVE OUTCOMES.

Cohort	Policy	Com-plete	Frozen	Resol-ving	Action/objective evidence	PLV evidence
Coffee/StackThree	BC-RNN	Yes	Yes	Yes	objective strongest	failed vs. baseline
Eight-task breadth	BC-RNN	Yes	Yes	Yes	objective strongest	failed vs. baseline
Action-chunk breadth	DP	Yes	Yes	Yes	open-loop strongest	below Action L1; gate not met
MimicGen selected	DP	No	Yes	Yes	not tested	descriptive
Retained recurrent	BC-RNN	No	Yes	Yes	not tested	descriptive
Unseen family	ACT	Yes	Yes	Yes	action signal	failed confirmation
Domain variants	ACT	Yes	Yes	Yes	not tested	failed gate
Unseen semantic family	ACT	Yes	Yes	Yes	not tested	failed gate
Strict RoboCasa	ACT	Yes	Yes	Mixed	not promoted	failed gates
Candidate-only	ACT	No	Yes	No	N/D	N/D

common-seed plan distances; these are not completed manuscript results.

APPENDIX G
SAME-START AVAILABILITY AND DEVIATION LOG

The audit includes all 47 locally available nonzero-epoch binaries in nine retained task–dataset–seed grids: 6,016 observations (47×32 starts $\times 4$ policy seeds), no duplicate checkpoint/start/seed keys, and success-independent inclusion. Runs contain five or six checkpoints; conditional residuals are finite in six of nine runs.

Launch configuration was reconstructed during execution from the running job's launch arguments and the fixed experiment queue. After launch, the aggregation gate was tightened to at least eight finite runs. Policy inference, dynamics training, simulator stepping, and

raw row generation were unchanged. The final gate is therefore labeled post-hoc rather than preregistered.

Two raw fidelity/ordering gates pass; the teacher-corrected gate fails. Task-balanced excess-success correlation is 0.068, model–simulator excess agreement 0.096, and residual association after Action L1 and teacher replan −0.309 over six eligible runs. Descriptive Regret@30 is 0.017 for the excess proxy versus 0.031 for Action L1. Only two task differences are nonzero (exact two-sided Wilcoxon $p = 0.500$), insufficient for automatic reranking.

Task-balanced normalized-state MSE on held-out demonstrations is 0.0017 at one step and 0.0045 averaged over five demonstration steps. In the same-start audit, one-step policy-induced MSE is 0.063; final imagined–simulator MSE after 16 actions is 3.135. Raw model–simulator replan ordering is 0.519, versus 0.096 for teacher-corrected excess. Simulator-induced replan has task-balanced $\rho = -0.212$: learned-model error is not the only bottleneck.

The 504-checkpoint audit does not support sparse failures being hidden by averaging. Direct mean replan has $\rho = 0.557$, versus p90 0.448 and worst-20% CVaR 0.418; recursive values are 0.551, 0.498, and 0.459.

Appendix H
Controlled Stride and Capacity Check

Every condition in the controlled stride and capacity audit uses the same 47 retained checkpoints, 32 validation starts from the common 16-action candidate set, and four common diffusion seeds. Valid replanning contrasts are stride four versus eight at horizon 16 and horizon eight versus 16 within stride four. Stride eight, horizon eight has one policy call and is a fidelity-only control.

At deployed stride and horizon, width 512 changes oriented model-replan rho by −0.085 versus width 256; only two of nine paired runs improve. The task-bootstrap interval is $[-0.245, 0.142]$, exact paired Wilcoxon $p = 0.625$ (minimum 0.125 for four nonzero task means). Lower MSE establishes no ranking gain under the same eight-epoch budget.

Appendix I
Budget Sensitivity and Outcome-Noise Scale

Table V uses task-balanced aggregation and 10,000-draw randomization at top one and 10%/30% retained budgets. Top one chooses a single checkpoint; the other budgets retain candidate sets. PLV-1 has lower regret than Action L1 only at top one and is never best. At 30%, both lie inside the random interval; open-loop smoothness is below it.

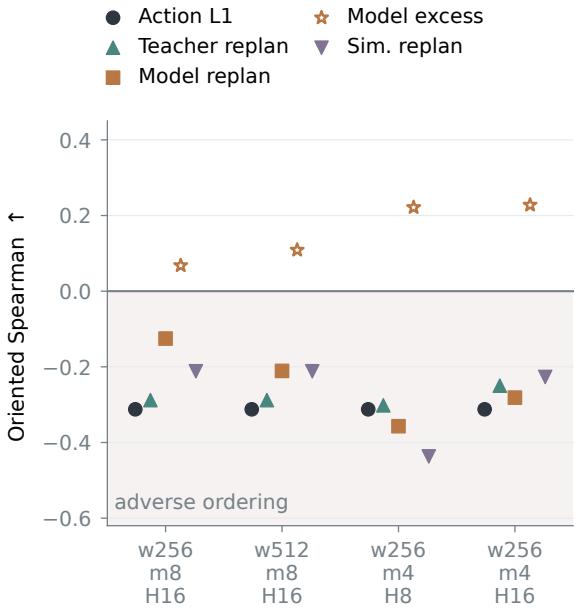


Fig. 1. Capacity and stride do not repair model-replan ordering. Points are task-balanced over nine same-start runs (47 checkpoints, four tasks). Negated costs make higher correlation better; shading marks adverse ordering. All replan branches remain nonpositive. Hollow stars denote derived teacher-corrected excess proxies, not measured branch scores. Evidence grade: mechanism audit.

With conditional split-half Spearman $r_{25} = 0.500$, Spearman–Brown reliability is $r_{50} = 2r_{25}/(1 + r_{25}) = 0.667$. Assuming noiseless metrics gives the one-sided classical attenuation approximation $\rho_{\text{true}} \approx \rho_{\text{observed}}/\sqrt{r_{50}}$. Action-L1, PLV-1, and objective-loss values become approximately 0.512, 0.532, and 0.055. This is neither a rank-specific correction nor a true-performance ceiling: the split conditions on archived success counts, omitting unarchived initial-condition or environment-seed variation.

Appendix J
Interpreting Screening Results by Regime

These regime interpretations are not deployment rules: the screening cascade still requires an informative frozen prospective allocation trial.

a) Strict prospective chronology: Each of two cohorts registered datasets, source-model hashes, environment, target tasks, and gate before target training, computed outcome-blind metrics for 126 checkpoints, and

TABLE IV

CONTROLLED SENSITIVITY, TASK-BALANCED OVER NINE RETAINED RUNS. NEGATED REPLAN COSTS MAKE HIGHER ρ BETTER; MODEL-REPLAN CORRELATION AND FIDELITY MSE APPEAR ONLY WHERE DEFINED.

Condition	Replan ρ	Final MSE	Reading
m=8, H=8, w=256	–	0.408	fidelity only
m=4, H=8, w=256	-0.357	0.557	shorter horizon
m=4, H=16, w=256	-0.281	3.270	stride check
m=8, H=16, w=256	-0.125	3.135	baseline
m=8, H=16, w=512	-0.211	1.309	lower MSE, worse rank

TABLE V

TASK-BALANCED REGRET (LOWER IS BETTER). BRACKETS GIVE THE RANDOM-SELECTOR 95% INTERVAL SHARED WITHIN EACH BUDGET.

Selector	R@1	R@10%	R@30%
Random (mean)	.207	.089	.046
Random (95% interval)	[.158, .266]	[.065, .112]	[.026, .064]
Action L1	0.163	0.067	0.033
PLV-1 smoothness	0.159	0.075	0.045
Open-loop smoothness	0.150	0.063	0.019

TABLE VI

INTERPRETATION BY REGIME, NOT A PROSPECTIVELY VALIDATED CASCADE. LATER STAGES DO NOT AUTOMATICALLY REPLACE EARLIER RANKINGS.

Regime	Primary signal	Decision
Heterogeneous pool	Action L1	default filtering
Matched BC-RNN	objective loss + Action L1	validate locally
DP near-tie (unvalidated)	teacher-forced replan	flag temporal disagreement
Hard/contact DP	boundary + overlap	discovery-only flag
Material disagreement	real rollout	abstain from reranking
New family / flat outcomes	recalibrate	no transfer claim

froze scores before 6,300 binary rollout episodes. No target outcome was used to fit, select, or recalibrate the scorer. Mean Spearman moves from 0.681 to 0.629 on the domain-variant cohort (Regret@20 0.010 $\rightarrow$ 0.017, Regret@30 0.007 $\rightarrow$ 0.017, 4/6 pools stable) and from 0.304 to 0.279 on the unseen bimanual dexterous family (Regret@20 0.050 $\rightarrow$ 0.050, Regret@30 0.047 $\rightarrow$ 0.050, 5/6 stable). Both fail their complete preregistered gates. The semantic-family cohort was the registered decisive generalization test; the domain-variant cohort is complementary, not a replacement.

APPENDIX K
WHERE THE DISCRIMINATIVE SIGNAL SITS

Across five metrics, worst-to-middle quintile gains span 0.133–0.273 in task-balanced success; the top 60% spans only −0.062–0.026. Top-quintile resolution, unlike global ρ, follows Regret@30 ordering (descriptive, not a test). The run-best success still exceeds each metric's best-quintile mean by about 0.14.

Within runs, action-space and smoothness scores track epoch at ρ = 0.56–0.69, success at 0.47. Yet selecting the last checkpoint performs poorly: success

TABLE VII

PRIMARY TASK-BALANCED SUCCESS BY WITHIN-RUN SCORE QUINTILE (Q1 WORST, Q5 BEST; Q2/Q4 OMITTED). Q1–Q3 IMPROVEMENT IS LARGER THAN Q3–Q5 IMPROVEMENT, DISTINGUISHING FULL-POOL RANKING FROM TOP-CANDIDATE CHOICE.

Metric	Q1	Q3	Q5	Q1→Q3	Q3→Q5
Open-loop smoothness	0.277	0.509	0.535	+0.232	+0.026
Action L1	0.252	0.517	0.531	+0.265	+0.014
PLV-1 smoothness	0.254	0.524	0.532	+0.270	+0.008
PLV-1 uncertainty+smooth	0.257	0.531	0.524	+0.273	-0.006
Objective validation loss	0.375	0.508	0.445	+0.133	-0.062

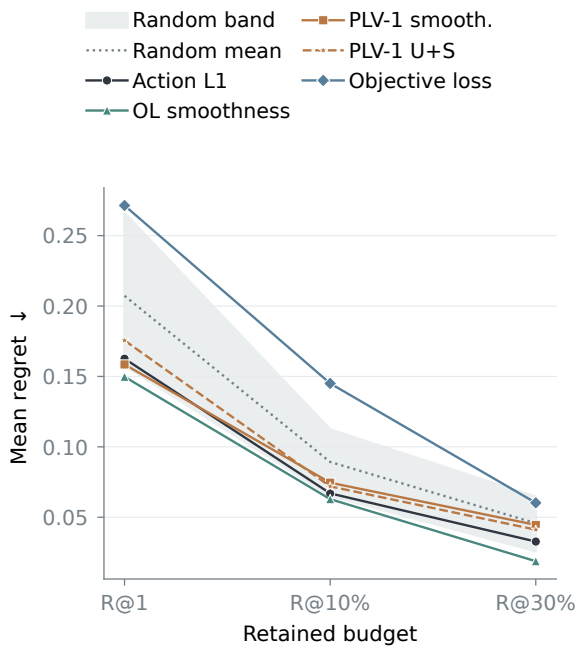


Fig. 2. Budget sweep over 45 runs and five tasks; lower regret is better. The band shows the task-balanced random-selector 95% interval (10,000 draws), the dotted line its mean. Only open-loop smoothness is below the band at all three budgets. Action L1 and PLV-1 track closely; objective loss is above the band at R@1/R@10% and inside at R@30%. Evidence grade: primary.

plateaus then declines late. Cheap metrics help avoid that decline more than rank survivors. Objective loss instead correlates with epoch at −0.353, tracking objective overfitting decoupled from deployed success and adverse on Diffusion Policy.

APPENDIX L
HOW MUCH IS LEFT TO WIN

Binomial decomposition estimates underlying SD 0.042 and headroom 0.063, versus observed gap 0.110. The 50-episode winner matches the underlying best with probability 0.322; near-ties limit expected selection loss to 0.012. Estimated SD is 0.072 on Transport/Tool Hang versus 0.009 on Can/Square.

Regret@30 uses a 50-episode maximum, mixing ability and sampling noise: its absolute scale is not all recoverable performance. Retained Can and Square candidates are nearly interchangeable, leaving little selection headroom; Transport and Tool Hang offer more, where execution-matched discovery was strongest.

APPENDIX M
EVIDENCE GRADES

a) Confirmatory support: Metric and endpoint were frozen before outcomes, the grid is complete, the cohort is outcome-resolving, and the declared comparison is met.

b) Descriptive support: Positive results use selected, incomplete, availability-limited, or metric-development coverage.

c) Failed confirmation: A frozen claim or promotion gate fails on complete, outcome-resolving coverage.

d) Non-discriminative: The prespecified outcome-variation criterion fails; neither positive nor negative selector evidence is assigned.

Selected MimicGen and retained recurrent results are descriptive. Coffee/StackThree, eight-task BC-RNN, action-chunk breadth, and unseen ACT use complete grids. Strict RoboCasa has frozen promotion gates but mixed outcome resolution; candidate-only RoboCasa is non-discriminative.

TABLE VIII

ABILITY-SPREAD ESTIMATE WITHIN ACTION L1'S RETAINED TOP 30%. OBSERVED GAP: SAMPLED MAXIMUM MINUS MEAN. BINOMIAL DECOMPOSITION ESTIMATES UNDERLYING SD AND BEST-MINUS-MEAN HEADROOM; HIT RATE ESTIMATES HOW OFTEN THE 50-EPISODE WINNER IS THE UNDERLYING BEST. PRIMARY RETROSPECTIVE EVIDENCE.

Task	Observed gap	True SD	Headroom	Hit rate
Can	0.046	0.008	0.013	0.162
Lift	0.090	0.049	0.070	0.420
Square	0.077	0.011	0.018	0.160
Transport	0.141	0.062	0.092	0.366
Tool Hang	0.198	0.083	0.120	0.501
Task-balanced	0.110	0.042	0.063	0.322

APPENDIX N
ARTIFACT MAP

The submission's `anc/results/` contains frozen manifests, run/pair summaries, bootstrap results, raw same-start rows, score-resolution and fidelity-reuse Parquet tables, gates, and historical checksums; `anc/code/` supplies selected generation, audit, and evaluator scripts. Each new Parquet table retains adjacent metadata JSON with input/output hashes, random seed, and Git commit. `anc/README.md` describes the inventory; `anc/SHA256SUMS` verifies submission bytes. Historical workspace paths require remapping; this is not a complete training repository. The cost–regret plot is omitted because runtimes lack a common hardware, batching, and reset benchmark.